

Toward Trust-by-Design in Federated Learning for Healthcare

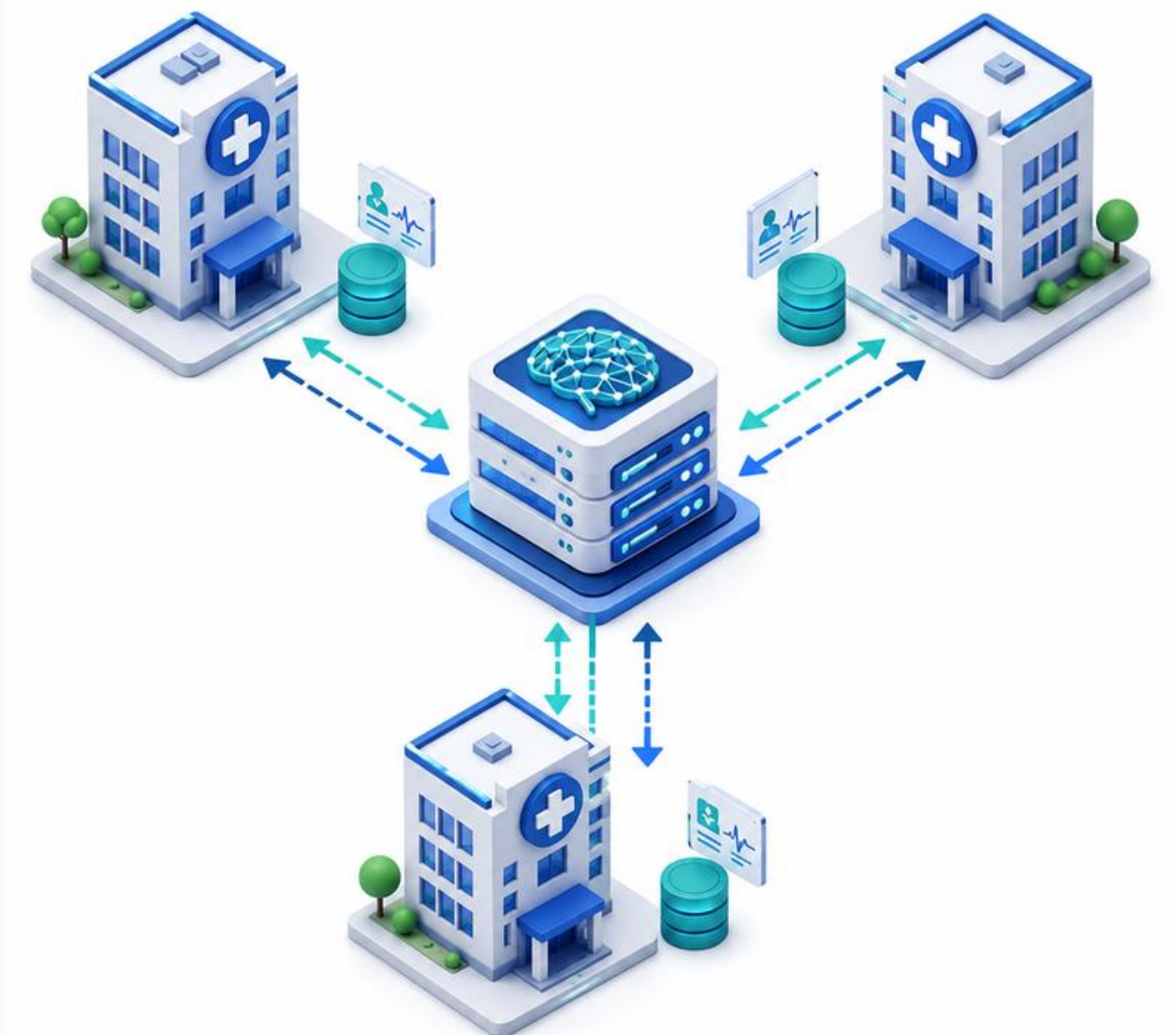
A Taxonomic Review of Privacy, Explainability, and Accountability

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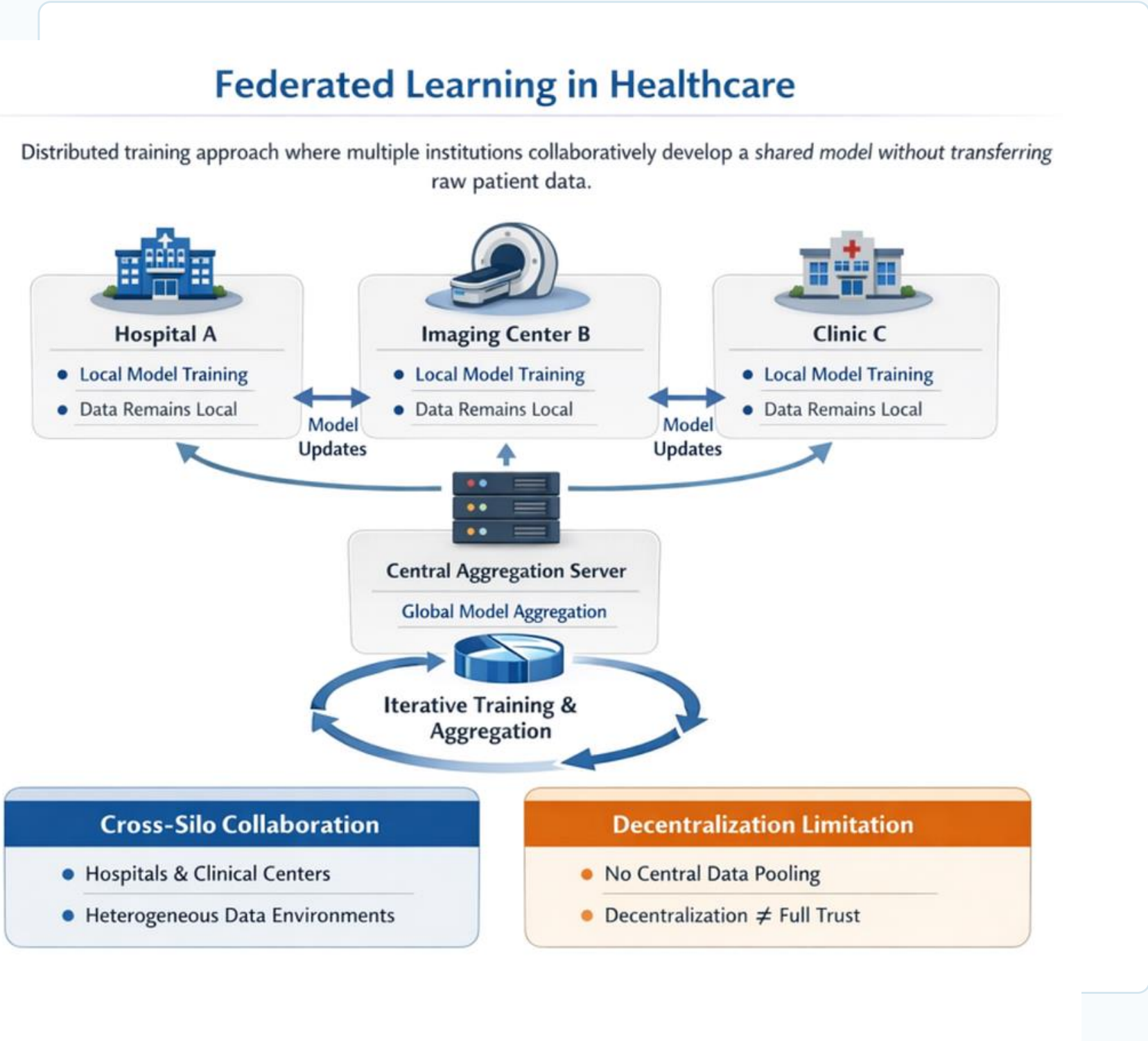


Na'mah Rassam Center
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Research Problem, Gap, and Review Objectives

Why conventional FL is necessary but insufficient for clinical AI



Research Problem

FL faces leakage risks, security vulnerabilities, weak explainability, limited accountability, and scarce deployment evidence.

Research Gap

Privacy, explainability, and accountability are usually studied separately rather than as one trust system.

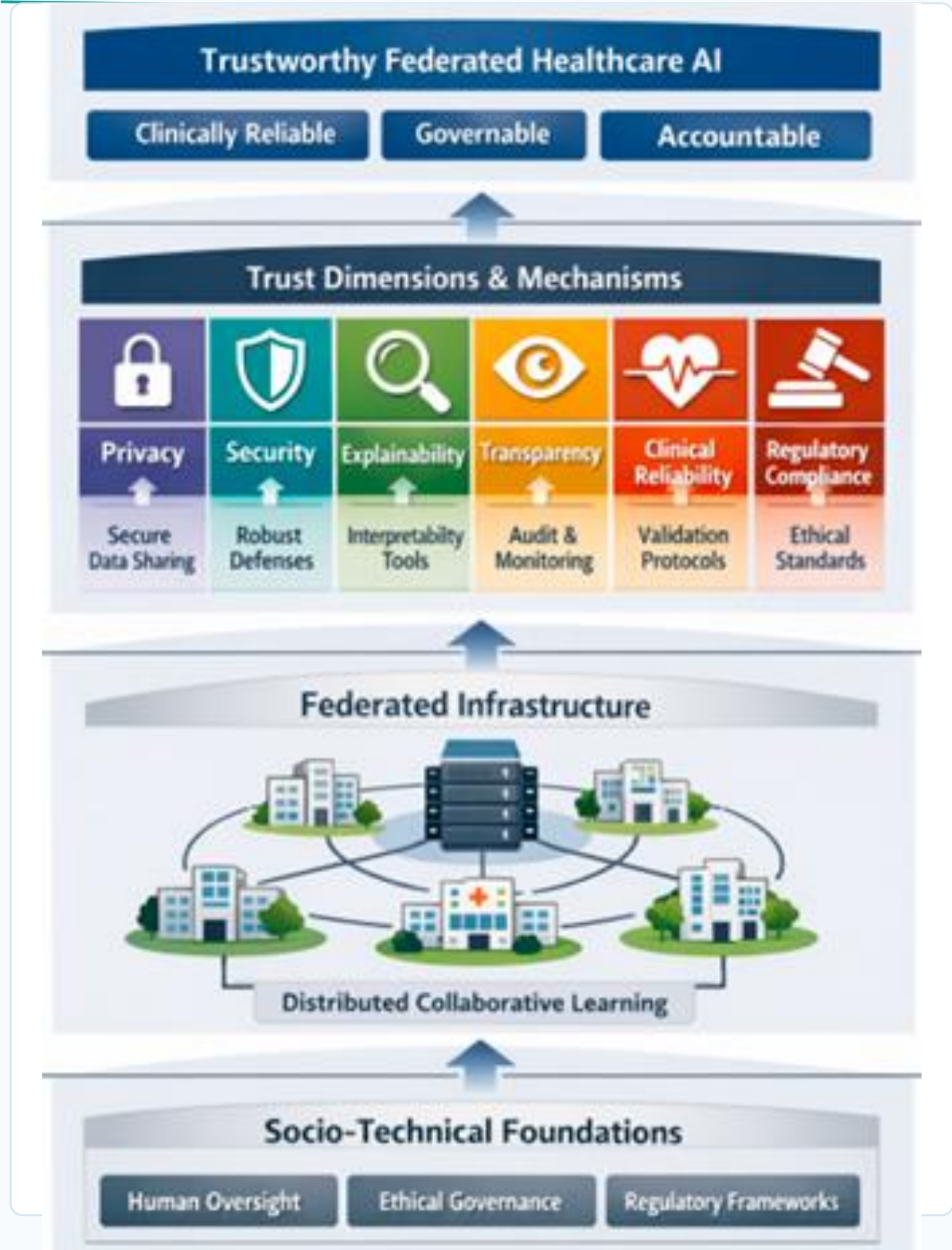
Review Objective

Develop a trust-oriented taxonomy, compare paradigms, evaluate evidence strength, and motivate Trust-by-Design.

Key reframing: the question is not only whether data can remain local, but how distributed model development can become trustworthy by design.

Core Trust Dimensions in Federated Healthcare AI

Trust is a socio-technical condition, not a direct output of decentralization



Privacy

Prevent leakage from raw data, gradients, and model updates.

Security

Resist adversarial manipulation and update-level attacks.

Explainability

Make model behavior understandable to clinicians.

Auditability

Enable inspection of development and update histories.

Clinical Dependability

Support safe decisions and workflow compatibility.

Governance

Align with ethical, regulatory, and institutional duties.

Design implication: trust must be engineered across technical safeguards, clinical oversight, and governance.

Taxonomic Framework for Trust-Enhanced FL

Table I: intervention locus × nature of trust assurance

TABLE I A TAXONOMIC FRAMEWORK FOR TRUST-ENHANCED FL IN HEALTHCARE					
Paradigm	Locus of Intervention	Nature of Trust Assurance	Representative Studies	Core Trust Claim	Fundamental Limitation
Foundational Paradigm: Architectural Privacy					
Architectural Privacy	Client	<i>Ex negativo</i> (trust by constraint)	[19], [26], [27]	Data never centralized; exposure precluded by system design	No formal guarantees against inference attacks; trust assumptions implicit and unverifiable
Cryptographic Paradigm: Mathematical Guarantees					
Communication-Channel Encryption	Communication	<i>Ex ante</i> (trust by secrecy)	[15], [28]	Mathematical infeasibility of information extraction from intercepted updates	Substantial computational overhead; assumes trusted key infrastructure
Aggregation-Level Differential Privacy	Aggregation	<i>Ex post</i> (trust by proof)	[13], [21], [29]	Quantifiable (ϵ, δ) bounds against inference attacks; formal privacy accounting	Privacy-utility trade-off; assumes trusted aggregator; narrow attack coverage
Hybrid Cryptographic	Multi-locus	<i>Composite</i> (trust by layering)	[20]	Combined guarantees across communication and aggregation layers	Highest cryptographic complexity; additive overhead; integration not formally characterized
Interpretive Paradigm: Transparency-as-Trust					
Visual Plausibility	Client	<i>Phenomenological</i> (trust by resemblance)	[11], [16], [22], [23], [30]	Explanations visually align with clinical expectations; attention maps highlight plausible regions	Plausibility conflated with correctness; no causal fidelity guarantees; clinician validation absent
Feature-Attribution	Client	<i>Reductionist</i> (trust by decomposition)	[31]	Model decisions decomposed into contributory feature weights; clinical predictors identified	Attribution does not imply causation; no validation of explanation utility in clinical tasks
Biologically Grounded	Client	<i>Evidentiary</i> (trust by mechanism)	[12]	Explanations link to quantifiable biological features; cross-center feature stability demonstrated	Analytical validation only; no clinician-in-the-loop assessment; mechanistic claims unverified
Accountability Paradigm: Verifiable Collaboration					
Update Traceability and Auditability	Aggregation	<i>Forensic</i> (trust by audit)	[14]	Blockchain-assisted coordination supports post-hoc traceability of update flow, system events, and collaborative model management	Sparse evidence base; simulation-bound; limited integration with formal privacy and interpretability
Unoccupied Paradigm: Integrated Trust-by-Design					
<i>Theoretically Possible but Empirically Vacant</i>	Multi-locus	<i>Emergent</i> (trust by co-design)	None	Simultaneous formal privacy guarantees, clinically validated explainability, system accountability, and deployment readiness	Within the reviewed studies, no paper was found to integrate all dimensions; represents the central research gap

Explainability Strategies in FL-Based Clinical Models

Table II: interpretive mechanisms and performance implications

TABLE II EXPLAINABILITY STRATEGIES AND THEIR PERFORMANCE IMPLICATIONS IN FL-BASED CLINICAL MODELS					
Study	Explainability Strategy	Representative Techniques	Performance Gain / Retention	Integration Level in FL	Effect on Model Performance
[22]	Post-hoc heatmap-based visual explanation	Grad-CAM	Competitive federated performance retained (86.5–87.1% accuracy)	Low to moderate	Performance retention alongside improved visual interpretability
[23]	Local surrogate and superpixel-based explanation	LIME	Strong federated performance retained (92.08% on ISBI2016, 94% on ISBI2017)	Moderate	Preserved model utility with enhanced local interpretability
[16]	Hybrid heatmap- and saliency-based visual explanation	Grad-CAM + Saliency maps	Good federated performance retained (approx. 94% accuracy)	Moderate	Preserved predictive performance and improved qualitative interpretability
[11]	Post-hoc heatmap-based visual explanation	Grad-CAM	Acceptable performance retained (88% accuracy; 89.66% F1-score)	Low to moderate	Enhanced transparency with retained predictive utility
[12]	Biologically grounded quantitative explanation	Integrated Gradients + quantitative cellular feature analysis	Cross-center predictive performance with analytically grounded biological explanation (overall AUROC 0.82)	High	Explainability linked to analytical validation and cross-center robustness
[30]	Multi-method local visual explanation pipeline	LIME + Grad-CAM	Improved pneumonia-prediction performance (97.61% accuracy, 98.36% F1-score)	Moderate to high	Strong overall performance, but independent contribution of each XAI component not quantified
[31]	Feature-attribution explainability for structured clinical FL	SHAP	Strong performance retained for CAD prediction (83.21% with privacy, 84.49% without)	High	Improved feature-level transparency with modest performance decline under privacy-preserving settings

Visual explanations retain model utility, but clinical usefulness remains undervalidated.

Privacy Strategies and Multi-Dimensional Trade-Offs

Table III: privacy protection, scalability, robustness, and deployment feasibility

TABLE III STRUCTURED PRIVACY STRATEGIES AND MULTI-DIMENSIONAL TRADE-OFFS IN FEDERATED CLINICAL LEARNING							
Study	Privacy strategy	Representative mechanisms	Perf. gain / retention	Privacy protection	Scalability	Attack robustness	Deployment feasibility
[19]	Privacy by decentralization	FL data locality, FedDEO + FedAvg	96.98% acc.; improved vs. baseline	Implicit / low-formality; no DP/HE	Moderate; simulated multi-client	Not explicitly evaluated	Limited; no real institutional evidence
[28]	Encrypted model-update protection	Homomorphic Encryption, FedAvg + IKMC	98.6% acc.	High at update/communication level; no formal budget	Limited to moderate; encryption adds cost	Partially addressed; leakage risk mitigated	Limited; simulated hospital setting
[29]	DP with secure aggregation	DP, Secure Aggregation, CNN-LSTM FL	Near-centralized utility retained (~1-2% decline)	High; explicit DP-based protection	Moderate; acceptable efficiency as devices increased	Partially addressed; model inversion targeted	Limited; fully simulated environment
[26]	Lightweight locality-based	Local data retention, weighted FedAvg	95.70% (3 clients) to 96.56% (5 clients)	Implicit / low-formality	Moderate to high; lightweight design	Not explicitly evaluated	Limited; simulated multi-center setting
[15]	Encrypted aggregation privacy	PHC, encrypted parameter aggregation	~10% avg. acc. improvement; 30% overhead reduction	High at aggregation level; no formal accounting	Moderate; PHC computation 150 ms	Conceptually addressed; leakage/misuse discussed	Limited; simulated collaboration
[27]	Decentralized privacy-aware	Data locality, adaptive FedAvg/FedSGD	Improved/stabilized learning under heterogeneity	Implicit to moderate; privacy via decentralization	Moderate; adaptive aggregation improves convergence	Conceptually addressed only; no formal testing	Limited; experimental framing
[21]	DP-based privacy hardening in SplitFed	DP noise, SplitFed architecture	Utility retained; primary gain was privacy hardening	Moderate to high; strong against label inference	Limited; time/space complexity underdeveloped	Strong for label inference specifically	Low to limited; preliminary interpretation
[13]	Formal DP-based	Gradient clipping, Gaussian noise, FL-DP	96.1% acc.; slight privacy-induced degradation	High (formal); $\epsilon = 1.9, \delta = 10^{-5}$	Limited to moderate; acknowledged overhead	Moderate to strong; membership inference addressed	Limited; experimental, no hospital-scale
[20]	Hybrid multi-layer	DP + HE, encrypted exchange, adaptive weighting	12% improvement over centralized baseline	Very high; multi-layer protection	Moderate; encryption adds nontrivial burden	Partially addressed; stronger protection	Moderate; among more translational, but complex

Stronger formal privacy usually increases computational and deployment burden.

Cross-Paradigm Trade-Offs

Table IV: why no single paradigm delivers comprehensive clinical trustworthiness

TABLE IV
CROSS-PARADIGM TRADE-OFFS IN TRUST-ENHANCED FL

Paradigm	Primary Trust Claim	Characteristic Strength	Characteristic Weakness	Operational Cost	Clinical Readiness
Architectural Privacy	Data never centralized	Minimal overhead; simple implementation	No formal guarantees; inference attack vulnerability	Low	Limited
Cryptographic Guarantees	Provable privacy bounds	Quantifiable, verifiable protection	High computational burden; trusted infrastructure assumptions	High	Simulation-bound
Interpretive Transparency	Decisions are understandable	Clinical plausibility; transparency	Validation absent; plausibility conflated with correctness	Low to moderate	Prototypical
System Accountability	Collaboration is traceable and auditable	Post-hoc verification; provenance logging	Sparse evidence base; limited preventive protection; simulation-bound	Moderate to high	Experimental
The Unoccupied Paradigm: Integrated Trust-by-Design					
Integrated Systems	Trust emerges from co-design	Simultaneous transparency, privacy, accountability, deployability	No existing implementations; interaction effects unknown	Unknown	None

Low cost
limited formal proof

Strong privacy
high operational burden

Clinical plausibility
validation gap

Traceability
sparse evidence

Evidence Quality and Validation Gaps

Tables V–VI: from technical evaluation to clinical deployment readiness

TABLE V
EVIDENCE STRENGTH ACROSS TAXONOMIC PARADIGMS

Study	Paradigm	Internal	External	Cross-Site	Clinician	Formal Privacy	Security Testing	Evidence Strength
[22]	Interpretive (Visual)	Yes	No	No	No	No	No	Limited
[19]	Architectural	Yes	Yes	No	No	No	No	Moderate
[28]	Cryptographic (Encryption)	Yes	Yes	Partial	No	No	No	Moderate
[12]	Interpretive (Biologically Grounded)	Yes	Yes	Yes (real)	No	No	No	Strong
[13]	Cryptographic (DP)	Yes	No	No	No	Yes	Partial	Moderate
[20]	Cryptographic (Hybrid)	Yes	Yes	Partial	No	Yes	Partial	Moderate
[21]	Cryptographic (DP)	Yes	No	No	No	Partial	Yes	Moderate
[15]	Cryptographic (Encryption)	Yes	No	Partial	No	No	No	Limited
[26]	Architectural	Yes	No	Partial	No	No	No	Moderate
[23]	Interpretive (Visual)	Yes	Yes	No	No	No	No	Moderate
[31]	Interpretive (Attribution)	Yes	No	No	No	Partial	No	Moderate

TABLE VI
EXPLAINABILITY VALIDATION ACROSS INTERPRETIVE PARADIGM

Study	Validation Method	Main Limitation
[22]	Qualitative visual inspection	No quantitative evaluation; no external validation
[23]	Qualitative LIME inspection	No clinician validation; no uncertainty modeling
[16]	Qualitative visual only	No external validation; no quantitative metrics
[11]	Visual verification of regions	No clinician validation; no stability analysis
[12]	Quantitative feature analysis	No clinician reader study; no uncertainty modeling
[31]	SHAP attribution	No clinician validation; simulated setting only

Validation Pattern

Evidence remains siloed: privacy studies emphasize formal guarantees, explainability studies emphasize plausibility, and clinician-in-the-loop validation is absent.

Deployment-ready evidence requires multi-site validation, attack testing, fidelity metrics, clinician usability, and operational benchmarking.

Future Research Directions

Integrated architectures

Co-design privacy, explainability, accountability, and readiness.

Clinical explainability

Validate explanations with fidelity metrics and clinician tasks.

Deployable privacy

Test privacy and auditability in realistic multi-center settings.

Conclusion

Taxonomic contribution

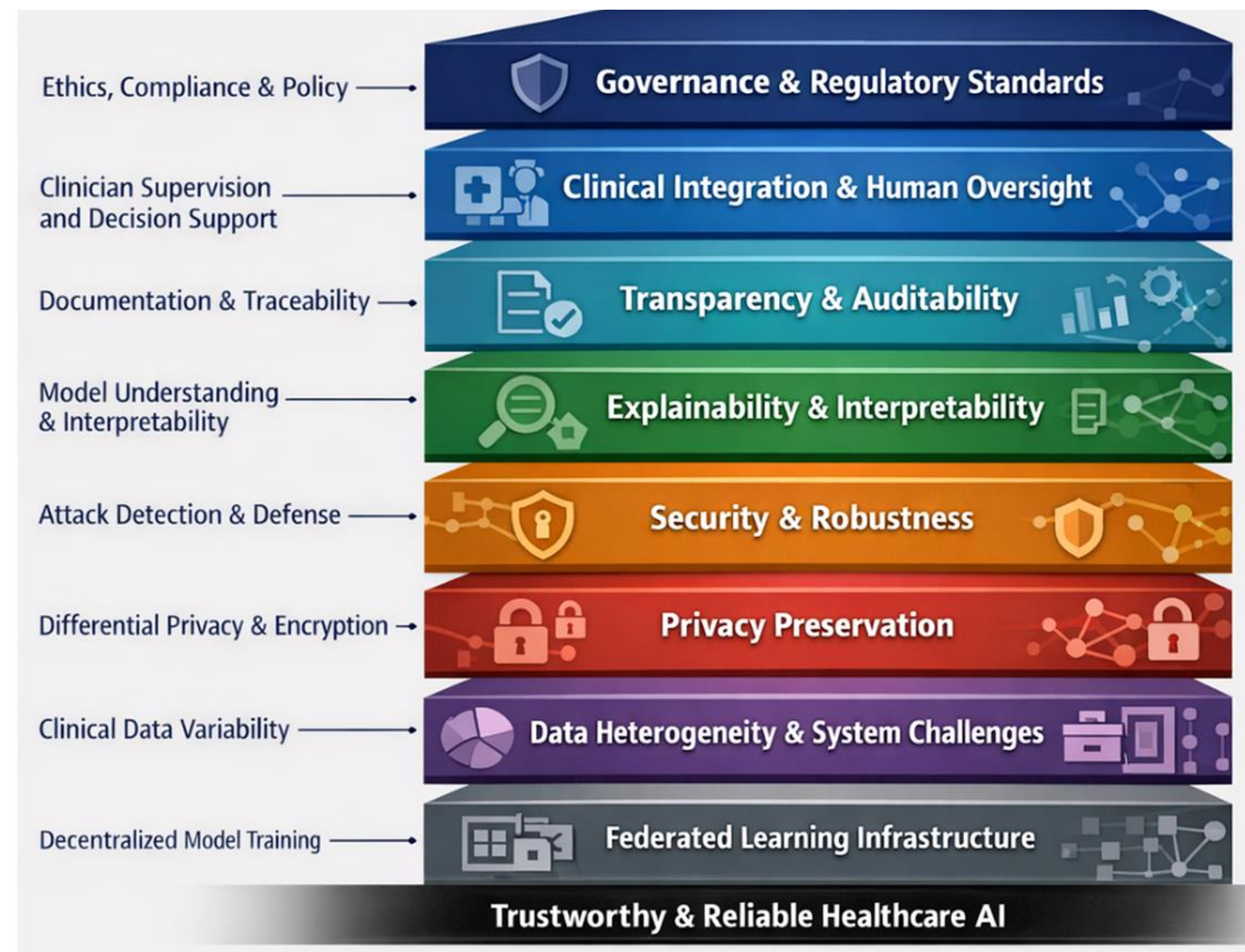
FL is reframed as trust engineering.

Core finding

Evidence remains paradigm-specific and fragmented.

Central gap

No reviewed system integrates privacy, explainability, accountability, and deployment readiness.



Thank You

Questions and
Discussion

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